WEB TECHNOLOGIES	
CSD 325	**CREDITS : 3**
INSTRUCTION : 2 Periods & 1 Tut/Week	**SESSIONAL MARKS : 40**
END EXAM : 3 Hours	**END EXAM MARKS : 60**

Prerequisites:

- Basic Knowledge of Programming Fundamentals
- Knowledge of Programming Languages (C,JAVA,SQL)

Course Objectives:

The course should enable the students:

- To learn designing of dynamic web pages by embedding with CSS, Java Script.
- To know how to design and develop simple database driven web application.
- Making the web applications using Node JS and Express JS
- To know how to design the web pages using Python
- Understanding with NoSQL database technology.

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Course Outcomes:

By the end of the course, the student will be able to

1.	Design the Webpages and templates.
2.	Demonstrate on creating a website using PHP and MySQL functionalities.
3.	Solve complex type of web applications using Node JS and Express JS
4.	Develop the Web applications using Python.
5.	Create the Document Oriented Database to interact with NoSQL

CO-PO Mapping:

CO	PO												PSO	
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**1**	**2**
1	2	2	2	1	2	1	-	-	2	1	2	2	2	-
2	2	2	2	2	2	-	-	-	2	1	2	2	2	-
3	2	2	3	2	2	-	-	-	2	1	2	2	2	-
4	2	2	2	2	2	-	-	-	2	1	2	2	2	-
5	2	2	2	2	2	-	-	-	2	2	2	2	2	-

SYLLABUS

UNIT 1: **16 Periods**

HTML: Basic Tags, fonts, hyperlink, lists, tables, images, frames forms, Cascading style sheets.
Java Script: Introduction, Objects, Events, DHTML.
Learning Outcomes: At the end of this unit the Students will be able

1. To design webpages and validation of webpages.
2. Test and debug JavaScript web applications.

UNIT 2: **12 Periods**

PHP: Introduction, Variables, Data types, control statements, Arrays, Functions, Strings, Form processing, File Handling, Cookies and Sessions, PHP with Database.
Learning Outcomes: At the end of this unit the Students will be able

1. To develop the PHP scripts to handle HTML forms.
2. To Create PHP programs that use various PHP library functions, and that manipulate files and directories.

UNIT 3: **12 Periods**

NodeJS: Basics and Setup, REPL terminal, NPM and Command Utilities, Global Objects, Modules – (OS, Path, DNS, Net, Domain),File System.
ExpressJS : Environment, Routing, Http Methods, URL Binding, Middleware, Templating, Static files, Form data.
Learning Outcomes: At the end of this unit the Students will be able

1. To Develop interactive Web applications.
2. To Develop single page applications.

UNIT 4 **12 Periods**

Flask: Introduction, Installation, Routing , URL Building, Http methods, Templates, Request Object, Sessions and cookies, File Uploading, Redirect & Errors, Flashing.
Learning Outcomes: At the end of this unit the Students will be able

1. To design webpages using python and validation of webpages.
2. Apply the routing techniques to access the desired pages.

UNIT 5: **12 Periods**
MongoDB: Introduction**,** Installation, Data Types. Data Modelling. **Database-**Create, Drop. **Collection-**Create, Drop. Aggregation, Indexing, CRUD-Operations,

Learning Outcomes: At the end of this unit the Students will be able

1. To learn how to store and access data from database and dynamic database activities.
2. To learn how to run queries against a MongoDB instance in order to store, manipulate, and retrieve data on it.

Text Books:

1. Thomas A. Powell, - “HTML & CSS Complete Reference”,Mc Graw Hill, Fifth Edition,2017.
2. Steven Holzner ,”Php: The Complete Reference “, McGrawHillEducation , fifth edition ,2008.
3. Reg Lim, “ Beginning Node.js, Express & MongoDB Development”, Kindle Edition, 2019.
4. Miquel Grinberg ,-Flask Web Development, O'Reilly,Second Edition,2018

Reference Books:

1. Kogent Learning Solutions Inc. Web Technologies: HTML, JAVASCRIPT , PHP , JAVA , JSP , ASP.NET, XML and Ajax,Black Book, Dreamtech Press , 1st edition (1 January 2009).
2. Kyle Banker, MongoDB in Action, Manning, Second Edition, March 2016.

Web Resources:

1. https://nptel.ac.in/courses/106/105/106105084/
2. https://www.coursera.org/learn/html-css-javascript-for-web-developers
3. https://www.coursera.org/learn/server-side-nodejs

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